

Gender Differentiation on the Determinants and Intensity of Adoption of Purdue Improved Cowpea Storage (Pics) Bags in Northern Nigeria

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Abstract

This study assessed gender differentiation on the determinants and intensity of adoption of Purdue improved cowpea storage (PICS) bags in northern Nigeria. Primary data was collected via a well-structured questionnaire from Kano, Kaduna, Gombe, Bauchi, Plateau, and the Niger States. Each State was purposively selected based on the concentration and availability of female PICS bag users. From the sampling frame of 2989, 20% of male and female small-scale cowpea farmers were randomly selected giving a total size of 598 cowpea farmers. Descriptive and double-hurdle regression models were used. The result of the socio-economic analysis indicated the mean age of farmers to be 42 years with an average of 8 persons per household and a dependency ratio of 1.19. Years of schooling were 10 with a farming experience of 25 years and 2.17ha as the average farm size. Results of the Logit model in males were significant for (extension contact, PICS information from other farmers and radio sources, and cowpea income, bags non-available) and for females (age, awareness, extension contact, PICS information from other farmers and radio source, bags non-available). The truncated regression model in males (cowpea income) and females (education, output, and bags non-available). Adoption of PICS bag technology by cowpea farming households was very instrumental in enhancing female farmers' adoption. It is therefore recommended that policymakers should put in place methods to motivate female farmers to further adopt this technology.

Keywords: Gender, Adoption, PICS technology, double hurdle, Nigeria.

1. Introduction

Purdue University partnered with the International Institute of Tropical Agriculture (IITA) to promote Purdue Improved Cowpea Storage" (PICS) bags for grain storage in several countries in West and Central Africa (Baributsa et al., 2016). This project was first developed in Cameroon, but it has since spread to twenty-three nations in Africa and Asia. This technology was created with the trademark PICS bags for Central and West African cowpea producers to prevent the destruction of cowpea by weevils. In Nigeria in 2008, PICS started with pilot activities in 100 villages. By the end of the 2009/2010 storage season, PICS bags had conducted a demonstration in 11,643 villages (Ibro et al., 2014).

Hermetic storage is an ancient technique that can use any sealed airtight container. Larry Murdock and colleagues put in place a technology triple layer pics bags that was reported cost-effective and scale neutral (Baributsa et al., 2010).

Nigeria like most sub-Saharan African countries is an agrarian economy. More than 70% of the population live in rural areas and are dependent on agriculture for a living. The livelihoods of most Nigerians and many other Africans depend on their ability to produce and market agricultural products (Ayanwale et al., 2013b).

Cowpea is one of the most important commercial and nutritional legumes grown by African farmers. Because of its high protein content affordable cost and increasing pop growth, there is an increasing demand for cowpea in this region.

According to Ayanwuyi and Akintonde (2012), cowpea provides food, and fodder and also improves soil fertility. It is the major source of plant protein, vitamins, and feeds for livestock; and because of this, it is often referred to as the poor man's meat. Cowpea is also an important trade commodity in Nigeria. It is ascertained that cowpea contains 25% protein and 64 % carbohydrate.

In African traditional society, men are usually considered breadwinners in households but it has been noticed that women play a key role in sustaining the family. Ibnouf, 2009 reported that women provide about 60 – 80 percent of the food in most developing countries which makes them responsible for half of the world's food production although their contribution is always overshadowed and attributed to men. Women usually embark on non-remunerated activities such as domestic work, care giving, daycare, preparation of meals, etc which undervalue their contribution and impact thereby under looking their impact Ajani (2008). However, women's contribution to agriculture in Nigeria is limited (Ibro et al., 2014). Therefore, the

impact of PICS bag adoption by gender will give an insight into women's contribution to adoption.

The objective of this paper is to determine the factors that would encourage the adoption and intensity of hermetic storage technology by gender.

2. Materials and Methods

2.1. Description of the Study Area

This study was conducted in three geopolitical zones of Northern Nigeria North East, North West, and North Central (Odiegwu et al. 2012). In each zone, two States were selected Gombe and Bauchi in the North East zone, Kano and Kaduna in the North West zone, and Plateau and Niger in the North Central zone (figure 1). Agro ecologically, Northern Nigeria is Sudan and Sahel savannah. Cowpea is known to be the staple food in the area.

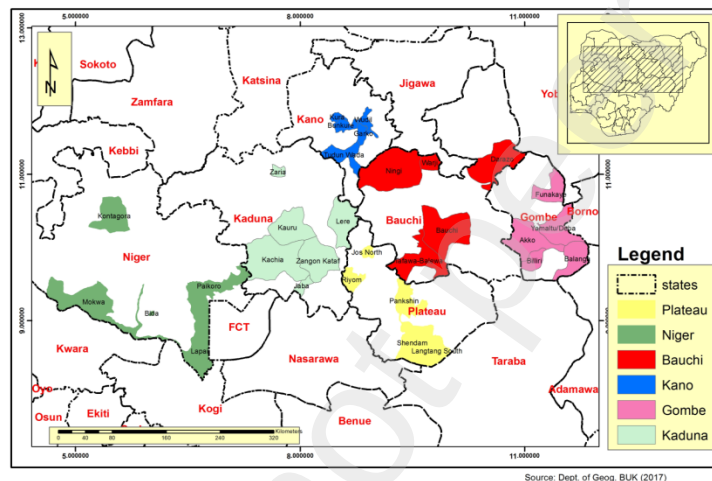


Figure 1: Map of the study area

2.2. Data Collection Methods

The data was gotten from household heads via a semi-structured questionnaire. Enumerators used the local language (Hausa) to administer the questionnaires. Recruitment and training of ten enumerators were done for two days on data collection methods and related topics. The researcher supervised the enumerators closely during the process. Respondents were told their information will be treated confidentiality, and research objectives were explained to them before the interview.

2.3. Sampling Procedure and Sample Size

The research was carried out in some selected states in Northern Nigeria because of the availability of PICS farmers. A combination of purposive and multistage sampling

procedures was employed for the study. The first stage in the sampling procedure involved the purposive selection of two states from each of the three agro-ecological zones of Northern Nigeria based on the concentration and availability of female cowpea farmers and PICS bag users. The states selected were Kano and Kaduna in North West, Gombe and Bauchi in North East as well as Plateau and Niger in North Central. Stage II involved the purposive selection of at least five (5) Local government areas (LGAs) from each state where there was widespread usage of PICS bags. Stage III involved a purposive selection of communities based on the abundance of female cowpea farmers. In the last stage, based on a list provided by the IITA Kano station, 20% of male and female small-scale cowpea farmers were randomly selected using numbers on Microsoft Office Excel. Participants in PICS training who have been using the bag for at least one year were classified as the treatment group while farmers who were non-participants in PICS demonstration/training and are not using the bag were classified as a control group.

These random selections resulted in a total of 299 PICS bag users comprising 138 men and 161 women. The respondents of the control group were selected from non-PICS villages (i.e villages where PICS demonstration/training was not carried out) from the same LGAs which are similar in characteristics to the villages where PICS demonstration was carried out. These villages were identified and randomly selected resulting in a total of 299 respondents including females and males. Therefore, the total sample size of respondents was 598 cowpea farmers.

2.4. Data Analysis Methods

Data were analyzed using descriptive statistics (percentage, frequency, mean), binary logistic, and Tobit regression models through Stata 15 software.

2.4.1. Double Hurdle Model (DHM) specification

The double hurdle model was used to measure factors influencing the probability of adoption and intensity of adoption of PICS bags. The double-hurdle model is a parametric generalization of the Tobit model as reported by (Greene, 2007, Asfaw et al. 2010, Cragg, 1971 and Awotide et al. 2014).

The double-hurdle model is applied in such a way that, both hurdles (the decision for adoption and intensity of adoption) have equations associated with them, incorporating the effects of the farmer's characteristics and circumstances.

The double-hurdle assumes that households make two sequential decisions about adopting and intensity of use of technology. Each hurdle is conditioned by the

household's socioeconomic characteristics. In the model, a different latent variable is used to model each decision process. The first decision variable (Z) takes the value 1 for farmers who have adopted PICS bags and take the value zero for otherwise.

The expected utility of adopting a technology (Z_i^*) is latent however. The first hurdle is the adoption decision equation estimated by using a Logit model given as: `

$$Z_i^* = X_{1i}'\beta_i + u_i \dots\dots\dots (1)$$

$$Z_i = 1 \text{ if } Z_i^* > 1$$

$$Z_i = 0 \text{ if } Z_i^* \leq 1$$

Where,

Z_i^* = latent variable that takes the value of 1 if the farmer adopts PICS bag and 0 otherwise, X_{1i}' = vector of explanatory variables (farmers/farm specific characteristics, institutional and technology specific characteristics) that influences adoption choice, β_i = vector of parameters and u_i = independently distributed error term.

The second hurdle of the double-hurdle model involves an outcome equation which uses a truncated model to determine the intensity of adoption measured in terms of the proportion of cowpea stored in PICS bags. Therefore, the second hurdle uses observations only from those cowpea farmers who indicated a positive value on the use of PICS bags. The truncated model is expressed as follows

$$Y_i^* = X_{2i}'\beta_2 + v_i \dots\dots\dots$$

(2)

$$Y_i = \begin{cases} Y_i^* & \text{if } Y_i^* > 0 \text{ and } D^* > 0 \\ 0 & \text{otherwise} \end{cases}$$

Where,

Y_i = observed response on the proportion of cowpea stored in PICS bags, X_i = vector of explanatory variables, β = vector of parameter estimates, v_i = error term.

The observed value of the proportion of cowpea stored in PICS bags is therefore expressed as:

$$Y_i = Z_i Y_i^* \dots\dots\dots$$

(3)

The error terms of the two decision models (adoption and intensity of adoption models) are distributed as follows:

$$u_i \sim N(0,1)$$

$$v_i \sim N(0, \delta^2) \dots\dots\dots (4)$$

Therefore the empirical model used to estimate the level and of intensity of adoption is given below:

$$Y_i = \beta_0 + \beta_1 X'_1 + \beta_2 X'_2 + \beta_3 X'_3 + \beta_4 X'_4 + \beta_5 X'_6 + \beta_7 X'_7 + \beta_8 X'_8 + \beta_9 X'_9 + \beta_{10} X'_{10} + \beta_{11} X'_{11} + \beta_{12} X'_{12} + e \dots\dots\dots$$

(5)

β_0 is constant, $\beta_1 - \beta_{12}$ is logistic regression coefficients and e is error term.

Table 1. Description of variables used in the logit regression model.

Variables	Description	Expected Effects
Dependent variable		
Adoption	1 if a respondent adopted PICS and zero if not.	
Quantity of cowpea stored in PICS Bags	Continuous (kg)	
Explanatory variables		
age	Continuous variable indicating the age of the respondent (years),	+
information about PICS bags from other farmers	1 if a respondent received information on PICS from other farmers and zero otherwise	+
household size	Continuous variable indicating the number of people in the household	+
awareness	1 if a respondent aware about PICS bags and zero otherwise	+
information about PICS bags from radio	1 if a respondent received information on PICS from radio and zero otherwise	+
education	Continuous variable indicating the years of schooling	+
contact with extension agent	1 if a respondent have contact with extension agent and zero otherwise	+
cowpea income	Continuous variable indicating the amount of cowpea sold	+
output	Continuous variable indicating the quantity of cowpea harvested	+
membership of association	1 if a respondent is a member of farmers' group and zero otherwise	+
used other methods to store cowpea	1 if a respondent uses other methods and zero otherwise	+
PICS bags available	1 if bags are available and zero otherwise	+

3. Results and Discussion

3.1. Characteristics of Farmers

The average age of a farmer was 43 and 41 years for female and male farmers respectively (Table 2). The average household size was approximately 8 persons per household for both female and male farmers. The dependency ratio was 1.08 and 1.19 for female and male responding households respectively. About 70% of both female and male responding households have at least primary education. However, the highest mean years of schooling was 10 years in female responding households whereas for males was 9 years of schooling. The average farming experience, however, did not vary widely among the groups as the variation was between 23 years for the female farmers and approximately 25 years for male farmers. The average for all the groups was approximately 24 years. The average farm size in the surveyed area ranges from 1.78 ha in female households to 2.17 ha in male households, thus showing minimal disparities (0.39) in farm sizes among farming households.

The percentage of extension visits was greater than 50 for female and male farming households respectively (Table 3). The average number of years spent in an association was five years for females and three years for males. Table 4 reveals that the majority (above 88%) of treatment (both female and male farmers) got information about PICS bags. The analysis showed that households that obtained credit for farming purposes were higher among female households in the surveyed areas. Overall, very few farm households (32.5%) were able to obtain credit in that year.

Table 2: Age, Household size, Dependency ratio, Years of schooling, Farming experience and Farm size

Variables	FEMALE		MALE		POOLED	
	Treatment	Control	Treatment	Control	Female	Male
Age(years)	44.3 (11.2)	41.4 (13)	42.2 (11)	39.5 (10.8)	42.8 (12.2)	40.9 (11.0)
Household size(number)	7.9 (4.4)	8 (4.9)	8.3 (4.8)	7.3 (4.2)	8 (4.5)	7.8 (4.5)
Dependency Ratio	1.07(1.29)	1.08(1.12)	1.21(0.88)	1.16(0.85)	1.08(1.2)	1.19(0.86)
Years of schooling	10.5 (5.6)	9.7 (5.7)	9.5 (5.3)	8.6 (5.7)	10.1 (5.6)	9.1 (5.5)
Farming Experience (years)	24.2 (12.8)	21 (12.7)	26.8 (12.4)	23.3 (12.9)	22.8 (12.8)	25 (12.7)
Farm size (hectare)	1.96(0.9)	1.59(0.71)	2.19(1.31)	2.16(1.03)	1.78(0.8)	2.17(1.18)

Values in parenthesis is standard deviations

Table 3: Contact with extension agent, membership of association, information on PICS and access to credit

Variables (percentages)	FEMALE		MALE		POOLED	
	Treatment	Control	Treatment	Control	Female	Male
N	161	161	138	138	322	276
Extension Contact	75.2	27.3	71.7	33.3	51.2	52.5
Membership of Association	72	70.2	74.6	61.6	71.1	68.1
Information on PICS	88.2	6.8	91.3	8	47.5	49.6
Access to credit	45.3	52.2	13	19.6	48.8	16.3

3.2 Determinants of Farmers Decision to Adopt PICS bags

The result of the estimated double-hurdle model is presented in Table 4. The double hurdle was justified because some of the factors that influenced the farmers' decision to adopt PICS bags were not the same factors that influenced the intensity of adoption of PICS bags by gender. To identify the factors influencing the decision to adopt PICS bags, the Logit model was estimated (first hurdle). The results reveal that six and five factors are significant in influencing farmers' decision to adopt PICS bags by female and male farmers respectively. These factors are the age of the farmer, awareness of the technology, access to extension contact, information about PICS bags, source of information from other farmers and radio, cowpea output, cowpea income, and bags non-available. The Log-Likelihood ratio (LR) of -71.69 and -58.97 of the estimated Logit model for Female and Male farmers respectively were significant at $p < 0.01$ and this indicates the joint significance of the explanatory variables included in the model.

The estimated coefficient of farmer's age in explaining the probability of adoption of PICS bags technology was positive and significant at $p < 0.05$ for female respondents. This is an indication that the age increment of farmers enhances their likelihood of adoption. This result is contrary to the findings of Awotide et al. (2014), who suggested that younger people are more likely to take up new technology than older farmers. The result of this study may be because aged farmers are likely to acquire knowledge and experience over time compared to younger farmers.

The coefficient of awareness was significant and positive for female farmers and pooled data at ($p < 0.10$) and at ($p < 0.05$) respectively. The result revealed that female households' awareness of PICS bags was higher than males. This indicates that awareness is a tool for PICS bag dissemination. Our study agrees with the findings of

Mekonen and Wubetie (2021) who reported that awareness was an important factor influencing individual behavior to adopt PICS technology.

The estimated coefficient of the visit of the extension agent has a positive and significant influence on the adoption of PICS bags for female, and male farmers and pooled data at ($p<0.01$), ($p<0.05$), and ($p<0.01$) respectively. This means that the likelihood of adoption tends to increase when a farmer has contact with an extension agent. A significant positive effect of extension agents on PICS technology was reported by Moussa et al. (2014) and Siyum et al. (2022) on improved bread wheat technologies.

The coefficient sources of information on PICS bags from other farmers and radio were found to be significant for both female and male farmers at $p<0.01$. The adoption of the PICS bags by female farmers has increased over time due to continued awareness. This is consistent with a key role played by farmers in the dissemination of information to their folks (Rabé, et al. 2021). The radio source was significant ($p<0.05$) and positive for male farmers only. The radio broadcast was a major source of information on PICS technology available to the farmers. The availability of reliable information sources will enhance the communication process with the adoption of improved technologies (PICS bags). This implies that male farmers have a higher and greater chance to listen to the radio broadcast. Radio is a useful source of information on improved agricultural technologies, Agwu et al. (2008).

The estimated coefficient of cowpea income was significant ($p<0.05$) and it was positive for male and pooled data. This implies as the income of the households increased by 1, the likelihood of using PICS bag technology increased by 0.004 factors. It suggests that as cowpea income increases, the probability of adoption of PICS bags increases as income makes it possible to meet the financial obligation in technology used. In this case, one can say that male farmers receive more income than female farmers. This corroborates with Mekonen and Wubetie (2021) and Bello et al. (2012).

The estimated coefficient of bags non-available was negatively significant ($p<0.01$) for female, and male farmers and pooled data. Everything being equal, it implies that the probability of PICS bags available increased by 2.67 for female and 3.57 for male farmers, as the availability of PICS bag supply at the harvesting time increased, farmers' use of PICS bags enhanced. Scarcity of PICS bag vendors, disconnection in information flow, and supply chain of PICS bags is the major constraint faced by

farmers in the study area. The lack of local availability of the bags was often the reason for not using PICS bags (Moussa et al. 2012)

The estimated coefficient for the output was positive and significantly ($p < 0.10$) influenced the farmer's decision to adopt female farmers. The results revealed that female-headed households were 55% more probable to use PICS bags than male-headed households. Farmers with higher output are likely to adopt the PICS bags thereby purchasing more bags to store their produce hence, increasing the probability of adoption. This corroborates with the findings of Mekonen and Wubetie (2021) and Ibro et al. 2014

Factors determining the extent of adoption of PICS bags were estimated using the truncated regression model as shown in Table 4. The results revealed some variation in the Logit and truncated regression models justifying the double hurdle. This implies that the factors that influenced the farmers' decision to adopt PICS bags were not the same factors that influenced the farmers' intensity of adoption. Education was positive and significantly ($p < 0.01$) related to adoption intensity for female farmers educated female farmers (17.5%) were more likely to use PICS bags than males. This implies that Education plays a critical role for farmers by enabling them to identify the problem of post-harvest loss and to change into practice the knowledge and skill acquired for the use and adoption of PICS. This is an inline result by Hassen et al. (2014).

The coefficient of the quantity of output store was positive and highly significant ($p < 0.01$) to adoption intensity for female farmers. This implies that female farmers had a 35.6% added advantage in terms of storage over male farmers. That is, the more the number of cowpea increases by the female farmer the higher the adoption intensity of PICS bags. This is consistent with Ibro et al. (2014).

Bags' non-availability was found to be significant to the adoption intensity of PICS bags at ($p < 0.01$) for female farmers. This implies that if bags are made available, the intensity of adoption of PICS bags will increase drastically in female-headed households by 43.8% more than in male households. This agrees with Ibro et al. (2014) who documented about farmers complained of bags not being available.

The coefficient of cowpea income was found to have a positive impact on PICS technology adoption for male farmers. This implies that male-headed households had a 20% increase in cowpea income than female-headed households. Households with a high cowpea income have a high probability to adopt PICS bags. This result

corroborates that of Mulaudzi and Oyekale (2015) who reported a variation in earnings from farming as a ratio of household monthly income.

4. Conclusion

Adoption of PICS bag technology by cowpea farming households was very instrumental in enhancing cowpea farmers who adopted the technology with females adopting more than male farmers. Several variables explained the adoption of the PICS technology including age, awareness, extension contact, information sources, radio source, and cowpea income and bags unavailable. The truncated regression model in males (cowpea income) and females (education, output, and bags non-available). Thus, there is a need to increase awareness and training of farmers and make the bags available across the study area at an affordable price. It is therefore recommended that policymakers should put in place methods to motivate female farmers to further adopt this technology.

Conflicts of Interest

The authors declare no conflicts of interest.

Acknowledgments

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Table 4: Double-hurdle estimates of determinants and intensity of adoption of PICS bags

Variables	FEMALE		MALE		POOLED	
	Logit model	Truncated model	Logit model	Truncated model	Logit model	Truncated model
Age(years)	0.05(2.4)**	1.8(0.95)	0.04(1.12)	13.6(0.53)	0.39(2.52)**	6.8(0.95)
Education(years)	0.26(0.49)	87.5(1.75)**	0.18(0.29)	414.9(0.69)	0.28(0.73)	271.1(1.36)
Household size(number)	-0.008(-0.18)	-6.3(-1.61)	-0.014(-0.20)	-51.7(-0.82)	-0.015(-0.39)	-23.6(-1.36)
Association(yes, no)	-0.006(-0.06)	-9.1(-0.21)	-0.037(-0.45)	-0.009(-0.45)	-0.03(0.42)	-8.1(-1.68)**
Awareness(yes, no)	1.43(1.66)*	-	1.03(1.29)	6.6(0.62)	1.16(2.04)**	3.9(0.61)
Extension contact(yes, no)	1.31(2.65)***	27.1(0.49)	1.36(2.56)**	21.1(0.36)	1.28(3.65)***	30.9(0.58)
Other farmers(yes, no)	3.62(5.75)***	-14.1(-0.22)	3.85(5.74)***	3.1(-0.77)	3.74(8.37)***	-5.5(-0.7)
Radio source(yes, no)	0.63(0.80)	-11.1(-0.22)	1.45(2.03)**	13.1(0.21)	1.08(2.10)**	-18.3(-0.54)
Cowpea Income(amount)	0.29(0.52)	0.0002(1.31)	1.45(2.27)**	0.004(2.0)**	0.94(2.33)**	0.001(3.19)***
Output(yes, no)	0.55(1.82)*	31.3(3.56)***	0.03(0.17)	44.8(1.27)	0.20(1.22)	39.8(2.77)***
Other methods (yes, no)	0.34(0.57)	19.6(0.28)	0.68(0.99)	17.6(0.99)	0.53(1.21)	17.9(0.87)
Bags non-available(yes, no)	-2.67(-3.67)***	-15.3(-4.38)***	-3.57(-3.13)***	-42.1(-1.04)	-3.02(-4.97)***	-39.7(-2.74)***
Constant	6.07(3.61)***	-123.9(-0.9)	5.39(3.21)***	-529.5(-0.92)	5.40(4.73)***	-929.4(-0.99)
N	322	157	276	134	598	291
LR Chi (12)	300.21	54.73	261.91	6.85	555.89	26.52
Prob>Chi ²	0.0000	0.000	0.0000	0.867	0.0000	0.0091
Pseudo R ²	0.6768		0.6895		0.6751	
Log likelihood	-71.69	-1012.9	-58.967	-924.72	-133.78	-1967.79
Sigma		207.7(11.4)***		969.8(3.46)***		602.6(6.86)***

***P<0.01, **P<0.05, *P<0.10, figures in parenthesis are z-values

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